

Haocheng Wang

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EDUCATION

Nanyang Technological University, Singapore — M.Computing in Applied AI (MCAAI) Oct 2025 – Oct 2026 (Expected)
GPA 4.63 / 5.0 | Coursework: Python · RL · Agentic AI · AI Algorithms · AI UX & Data Visualization · Data Governance

Chongqing Jiaotong University, China — B.Eng. in Architecture Sep 2011 – Jun 2017
Graduation with Honors [Top 15%] · Merit Student 2015 [Top 6%]

SELECTED PROJECTS

My Little Bedroom — RL Agent for Automated Furniture Layout [Live Demo](#) Apr 2026

- Built a custom Gymnasium environment and MaskablePPO agent for furnishing procedurally-generated bedrooms (13 furniture types, ~41K discrete actions per step). Designed a hybrid reward combining functional accessibility, perceptual privacy, and circulation clearance criteria drawn from architectural domain knowledge.
- Core finding: reward composition is itself a design language — multiplicative terms enforce non-negotiable quality floors, additive terms negotiate soft preferences. Trained agent generalizes across unseen room configurations, learning emergent spatial grammar (bed tucked from door, wardrobe to back wall) with zero demonstrations.

PRISM — Multi-Agent Paper Reading with Sentence-Level Citations [Live Demo](#) May 2026

- Built a 5-agent system on Google ADK + Gemini for arXiv paper analysis. Introduced a generate-then-locate citation pipeline: model answers from full paper context first, then pins each claim to its exact source sentence via cosine similarity + LLM rerank — avoiding the retrieval errors of classical pre-retrieval approaches.
- Designed sentence-level coordinate system (§.para.sent), 7-role discourse structure classifier with narrative spectrum visualization, 5-layer symmetric guardrails with honest-gap detection (13/13 probes pass), and a user-triggered self-correction loop (6-standard critique → regenerate).

Non-Intrusive Depression Screening — Knowledge Distillation Pipeline Nov 2025

- 3-phase progressive pipeline (algorithmic benchmarking → NN architecture search → Knowledge Distillation) recovering full-feature Teacher performance (F1 0.87, AUC 0.92) from only 10 objective behavioral markers (sleep, diet, study hours) — avoiding sensitive data collection entirely.
- Distilled student achieved +39% F1 and +60% recall over baseline on held-out test set. Latent space topology analysis revealed KD transfers the Teacher's decision geometry, not just its labels — acting as a strong regularizer against overfitting on weak-signal features.

Additional coursework: Census Income Fairness Audit — intersectional bias analysis, 9.74× exclusion gap, 3 mitigation strategies (CA6003) · London Energy Fingerprinting — K-Means on expert-engineered features, visual design lead (CA6002) · Traffic Forecasting — data-centric SimpleRNN, 35.6% RMSE reduction (CA6000)

PROFESSIONAL EXPERIENCE

Yanfei Architects, Shanghai — Project Architect · Class I Registered Architect (PRC) Jun 2018 – Dec 2024
2023 Nanjing Excellent Engineering Survey & Design Award, First Prize in Architecture

- Led end-to-end design of [Nanjing Xianlin School](#) (62,000 m²): multi-objective spatial optimization under complex regulatory constraints, coordinating cross-disciplinary teams (structural, MEP) to resolve spatial conflicts pre-construction.
- Engineered a modular furniture system for 40+ customized hotel rooms — decomposing bespoke requirements into standardized components for mass production, streamlining the design-to-fabrication workflow.

Atelier Li Xinggang, CADG, Beijing — Intern · Parametric Modeling · Project Research Oct 2016 – Jan 2017

PUBLICATION

[Soft Architectural Robot](#) — Design of a Multi-Layer Composite Material Apparatus with PAM Actuator and Its Architecture Implications. **Published in CAADRIA 2018.** Tsinghua University Parametric Design Workshop.

SKILLS

AI / ML: Python · PyTorch · TensorFlow / Keras · Gymnasium · scikit-learn · Google ADK · SQL

Methods: Reinforcement Learning · Knowledge Distillation · Multi-Agent Orchestration · Feature Engineering · Data-Centric AI · Fairness Auditing

Dev Tools: Claude Code · Cursor · Git · Microsoft Office Suite · Rhino + Grasshopper · ComfyUI · GIS (ArcGIS / QGIS) · Adobe Creative Suite

Languages: Mandarin (Native) · English (Professional Working Proficiency)